



Atomix: Your Personal Lab with AI

LY Innovation-Based Major Project-B Report

Submitted in partial fulfilment of the requirements of the Degree of

Bachelor of Technology in Computer Engineering

by

Kushal D. Soni (Roll No. [INSERT])

Kartik R. Verma (Roll No. [INSERT])

Dhir M. Thakar (Roll No. [INSERT])

Supervisor

Dr. Madhura Phadke



Department of Computer Engineering

K. J. Somaiya Institute of Technology

An Autonomous Institute permanently affiliated to University of Mumbai

Ayurvihar, Sion, Mumbai-400022

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CERTIFICATE

This is to certify that the project entitled "**Atomix: Your Personal Lab with AI**" is a bonafide work of **Kushal D. Soni, Kartik R. Verma, Dhir M. Thakar** submitted to the University of Mumbai in partial fulfilment of the requirement in Major Project, for the award of the degree of "Bachelor of Technology" in "Computer Engineering".

Dr. Madhura Phadke

Project Guide

Department of Computer Engineering

Dr. Sarita Ambadekar

Head, Department of Computer
Engineering

Dr. Vivek Sunnapwar

Principal, KJSIT

Place: Sion, Mumbai-400022

Date:

PROJECT APPROVAL FOR L. Y.

This project report entitled "**Atomix: Your Personal Lab with AI**" by

Kushal D. Soni (Roll No. [INSERT])

Kartik R. Verma (Roll No. [INSERT])

Dhir M. Thakar (Roll No. [INSERT])

is an approved Last Year Innovation-Based Major Project in Computer Engineering.

Examiners:

1. _____

Name and Signature

External Examiner

2. _____

Name and Signature

Internal Examiner

Place: Sion, Mumbai-400022

Date:

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Kushal D. Soni (Roll No. [INSERT])

Kartik R. Verma (Roll No. [INSERT])

Dhir M. Thakar (Roll No. [INSERT])

Date:

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Place: Sion, Mumbai-400022

Date:

ABSTRACT

Chemistry is learnt by linking what is seen in a laboratory to what happens at the molecular level and to the symbolic equation, yet students get little safe, repeatable practice and rarely receive specific explanations of their mistakes. Virtual laboratories, immersive Virtual Reality (VR), game-based learning and AI chatbots each address part of this problem, but they are usually studied in isolation, judge experiments only as fixed sequences, and offer help that is unaware of what the learner has just done. This project presents Atomix: Your Personal Lab with AI, a Unity 6 virtual chemistry laboratory that runs on Meta Quest / OpenXR headsets and, through a run-time-injected keyboard-and-mouse mode, on ordinary Windows computers. Atomix offers eight inorganic experiments selected from an in-world Reaction Book. A data-driven free-hand reaction engine judges each experiment by the order of addition, instant overdose detection, underdose after a settling period and heating procedure, using per-reaction tolerances scaled by Guided, Standard and Expert difficulty levels, and rates every success with an accuracy score and stars.

Before and after each experiment the learner can watch recorded molecular videos or a live 3D animation that shows bonds breaking and forming and electrons being transferred. A context-aware AI Lab Assistant receives the live measurements, verdict and attempt history; it answers by voice and text through the Convai service and falls back to a grounded offline knowledge base with local speech, so help remains available without internet or a microphone. A persistent experiment history, enthalpy, activation-energy, entropy and Gibbs-energy graphs, and a randomized assessment scene with eighteen task types, a coin economy and HTML/CSV reports complete the learning cycle. The implementation comprises 110 C# scripts (39,601 lines), compiles without errors and passed all automated checks executed on the shipped logic (111/111, 167/167, 226/226 and 22/22). A controlled study of learning outcomes is identified as the next step.

Keywords: virtual chemistry laboratory, virtual reality, conversational AI, game-based learning, Unity, free-hand experimentation.

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LIST OF ABBREVIATIONS

Sr. No.	Abbreviation	Description
1	AI	Artificial Intelligence
2	API	Application Programming Interface
3	CAVE	Cave Automatic Virtual Environment
4	CPK	Corey-Pauling-Koltun (atom colouring convention)
5	CSV	Comma-Separated Values
6	DFD	Data Flow Diagram
7	FR	Functional Requirement
8	GBL	Game-Based Learning
9	GPU	Graphics Processing Unit
10	HMD	Head-Mounted Display
11	HTML	HyperText Markup Language
12	HUD	Heads-Up Display
13	INR	Indian Rupee
14	JSON	JavaScript Object Notation
15	KB	Keyboard (in figures: KB + mouse) / Knowledge Base (Offline KB)
16	LOC	Lines of Code
17	MP4	MPEG-4 Part 14 video format
18	RAM	Random Access Memory
19	RCT	Randomized Controlled Trial
20	REST	Representational State Transfer
21	SDK	Software Development Kit
22	SO	ScriptableObject (Unity data asset)
23	SoC	System on Chip
24	SUS	System Usability Scale
25	TTS	Text-to-Speech
26	UI	User Interface
27	UML	Unified Modelling Language
28	VFX	Visual Effects
29	VR	Virtual Reality
30	XR	Extended Reality

CHAPTER 1: INTRODUCTION

1.1 Introduction

Chemistry is learnt at three linked levels: the macroscopic change seen in a test tube, the sub-microscopic world of atoms and bonds, and the symbolic level of formulas and equations [9]. Learners find the transitions between these levels hard because the molecular level is never directly observed [10], while practical laboratory time is limited by reagent cost, apparatus, supervision and safety rules that forbid reactions such as sodium in water. Technology-enhanced learning has responded with virtual laboratories [1], [12], game-based learning [2], [3], [11], immersive Virtual Reality (VR) [4]-[6], [13] and AI chatbots [7], [14]. This project, **Atomix: Your Personal Lab with AI**, is a Unity 6 virtual chemistry laboratory that runs on Meta Quest / OpenXR headsets and, without a headset, on a Windows desktop with keyboard and mouse. Figure 1.1 shows its learning cycle.

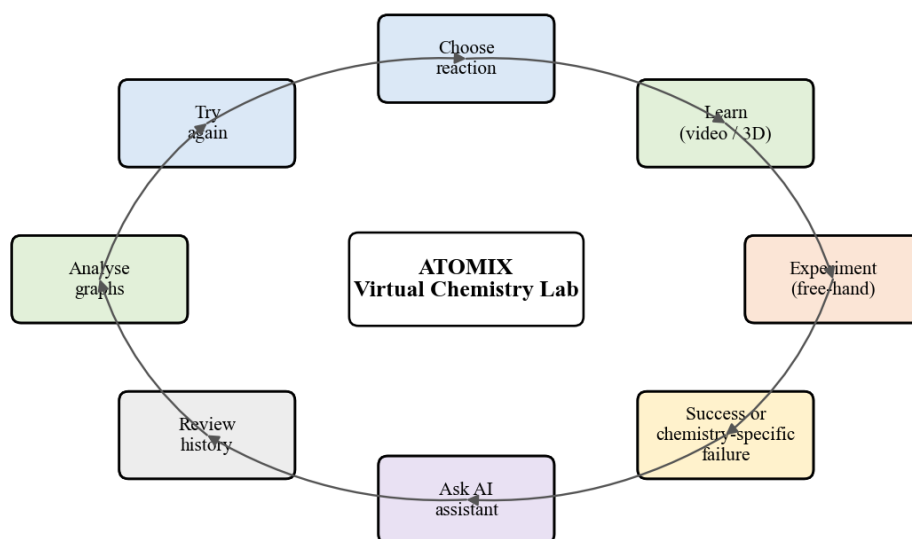


Figure 1.1: The Atomix learning cycle

1.2 Problem Definition

Students rarely get enough safe, repeatable practice with real reactions, and seldom receive immediate, specific explanations of their mistakes. Existing virtual laboratories judge experiments as fixed click sequences and are mostly desktop-only [1], [12]; VR applications offer immersion but do not assess the quantities used [4], [5], [13]; and AI

chatbots answer questions without knowing what the learner has just done, and stop working offline [7], [14]. The problem addressed is therefore **to build one accessible laboratory, usable in VR and on a desktop, in which learners perform quantity-aware experiments, see the molecular explanation, receive chemistry-specific feedback from an AI assistant that knows the live experiment and works offline, review their attempts and are assessed through randomized tasks.**

1.3 Aim and Objectives

Aim: to build Atomix, an AI-assisted virtual chemistry laboratory that unifies experimentation, molecular visualization, context-aware help, progress tracking and assessment for VR and desktop. **Objectives:**

1. Model an interactive 3D laboratory with eight inorganic reactions selectable from a world-space Reaction Book.
2. Support XR interaction on Meta Quest / OpenXR and a run-time-injected keyboard-and-mouse mode from one code base.
3. Build a data-driven free-hand engine that validates order, overdose, underdose and heating with configurable tolerances and difficulty levels.
4. Provide learn-before-perform molecular videos and a live 3D animation for all eight reactions.
5. Integrate a context-aware AI assistant (Convai) with an offline grounded knowledge base and speech.
6. Persist attempt history with accuracy scores and stars, and present thermodynamic graphs for each reaction.
7. Build a randomized 18-task assessment with a persistent coin economy and HTML/CSV reports, and verify the implementation with automated checks.

1.4 Organization of the Report

Chapter 2 reviews the literature; Chapter 3 specifies requirements; Chapter 4 presents the design; Chapter 5 the methodology; Chapter 6 the implementation; Chapter 7 the results; and Chapter 8 concludes.

CHAPTER 2: REVIEW OF LITERATURE

2.1 Literature Survey

Search strategy. IEEE Xplore, ScienceDirect, the Journal of Chemical Education and Google Scholar were searched for 2000-2024 using the keywords "virtual chemistry laboratory", "virtual reality chemistry education", "immersive molecular visualization", "game-based learning chemistry" and "AI chatbot education". Fourteen works on laboratory simulation, immersive or game-based chemistry learning, chemical representations and conversational AI were retained (Table 2.1).

Gilbert and Treagust [9] show that understanding chemistry depends on linking macroscopic, sub-microscopic and symbolic levels, and Wu et al. [10] found that visualization tools scaffold this link. Ramos et al. [1] argue that virtual laboratories are best used to complement and prepare for hands-on work, and Ali and Ullah [12] found wide variation among virtual laboratories in realism, feedback and assessment. Agarwal and Saha [2] and Sanchez-Lopez et al. [3] used puzzle games and gamified lab activities to raise engagement, and Byusa et al. [11] conclude that game-based learning improves motivation and understanding when mechanics serve learning goals; Kurniawan et al. [8] stress structured evaluation of such games. Bell and Fogler [4], Luo et al. [5] and Alvarez Ramirez and Bueno [13] applied VR and immersive visualization to chemistry, while Xie et al. [6] caution, from a randomized pilot, that immersion must be judged by measured outcomes. Sandu and Gide [7] and Atmosukarto et al. [14] report positive use of AI chatbots, but these are unaware of laboratory activity.

Table 2.1: Summary of the literature survey

Ref.	Author / Year	Method	Dataset / Participants	Results	Limitations
[1]	Ramos et al., 2016	Pedagogical integration of hands-on and virtual labs	Undergraduate chemistry course context	Virtual labs best used to complement and prepare for hands-on work	Desktop simulation; no immersive or AI support
[2]	Agarwal & Saha, 2011	Puzzle-based game (Atoms to Molecule)	Learners playing the game	Game format supported engagement with bonding concepts	Single concept; no laboratory procedure

Ref.	Author / Year	Method	Dataset / Participants	Results	Limitations
[3]	Sanchez-Lopez et al., 2023	Gamified lab activity (IQ-Mobile Race)	Chemistry laboratory class	Improved motivation, communication, collaboration	Physical activity; not a simulation
[4]	Bell & Fogler, 2004	VR modules for chemical-engineering education	Engineering students / VR modules	VR visualises processes not feasible in class	Early hardware; no quantity-level assessment
[5]	Luo et al., 2004	Immersive multi-projection molecular visualization	System demonstration	3D immersion aids understanding of molecular structure	Expensive projection setup; structure only
[6]	Xie, Li & Tang, 2023	Randomized controlled pilot study of immersive VR	Science learners (pilot RCT)	Immersion must be judged by learning outcomes	Pilot scale; VR content only
[7]	Sandu & Gide, 2019	Survey of AI-chatbot adoption	Indian higher-education students	Positive perception of chatbot support	Generic chatbot; no task context
[8]	Kurniawan et al., 2024	Systematic review of game-evaluation tools	Published evaluation studies	Need for structured usability and educational evaluation	Review only; no system
[9]	Gilbert & Treagust, 2009	Edited synthesis on multiple representations	Body of chemistry-education research	Macro / sub-micro / symbolic levels must be linked	Theoretical; no tool
[10]	Wu, Krajcik & Soloway, 2000	Computer-based visualization tool for representations	High-school chemistry classroom	Technology scaffolds representational understanding	2D desktop tool; no experimentation
[11]	Byusa et al., 2022	Systematic review of game-based learning in chemistry	Published GBL studies	GBL improves motivation and understanding	Effects depend on mechanics; heterogeneous studies
[12]	Ali & Ullah, 2020	Review and comparison of virtual chemistry labs	Existing virtual-lab products	Large variation in realism, feedback, assessment	Comparative review; no new system
[13]	Alvarez Ramirez & Bueno, 2020	VR application for organic chemistry	Prototype application	VR feasible for chemistry learning	Organic topics; no AI or assessment
[14]	Atmosukarto et al., 2021	AI chatbot in adaptive online chemistry course	Online course students	Chatbot extends support beyond class hours	Course-level; unaware of lab activity; online only

2.2 Research Gaps

Table 2.2 maps the surveyed work onto the capabilities a complete practice laboratory needs.

Table 2.2: Capability coverage of surveyed work versus Atomix

Capability	[1],[12]	[2],[3]	[4],[5],[13]	[6]	[7],[14]	Atomix
3D virtual laboratory procedure	Yes	-	Partial	Partial	-	Yes
Immersive VR	-	-	Yes	Yes	-	Yes
Desktop access without headset	Yes	Yes	-	-	Yes	Yes
Quantity / order-aware validation	-	-	-	-	-	Yes
Molecular-level visualization	-	Partial	Yes	-	-	Yes
Conversational AI help	-	-	-	-	Yes	Yes
AI aware of live experiment state	-	-	-	-	-	Yes
Works offline	Yes	Yes	Yes	Yes	-	Yes
Assessment and attempt history	Partial	Partial	-	Yes	-	Yes

The gaps are: (1) fragmentation - no surveyed system combines virtual lab, VR, molecular views, AI and assessment; (2) shallow validation - quantities, order and heating time are not judged, so realistic mistakes cannot be explained; (3) context-blind, online-only AI; (4) headset-only access; (5) a weak link between practice and thermodynamic theory; and (6) limited evaluation of educational effect. Atomix addresses gaps 1-5; gap 6 is future work.

CHAPTER 3: REQUIREMENT SPECIFICATION

3.1 Introduction

Requirements were derived from the problem definition, the research gaps and repeated play-testing. Estimated values are marked [ASSUMED].

3.2 Hardware Requirements

Table 3.1 lists hardware for development, desktop use and VR use. A microphone and internet are optional because questions can be typed and the assistant answers offline.

Table 3.1: Hardware requirements

Component	Development workstation	Desktop run-time (Windows)	VR run-time
Processor	Quad-core x64, 3.0 GHz or better [ASSUMED]	x64 dual/quad-core [ASSUMED]	Meta Quest on-board SoC, or PC as left
Memory	16 GB RAM [ASSUMED]	8 GB RAM [ASSUMED]	Headset built-in
Graphics	DirectX 11/12 GPU, VR-capable [ASSUMED]	DirectX 11 GPU	Headset built-in
Storage	50 GB free (project, Library, builds) [ASSUMED]	2-3 GB free [ASSUMED]	2-3 GB free [ASSUMED]
Display / HMD	Monitor + Meta Quest for testing	Monitor (any resolution)	Meta Quest / OpenXR headset
Input	Keyboard, mouse, Quest controllers	Keyboard and mouse	Quest controllers / hands
Audio	Speakers/headset, microphone	Speakers; microphone optional	Headset audio and microphone
Network	Internet for packages and Convai	Optional (cloud AI only)	Optional (cloud AI only)

3.3 Software Requirements

Table 3.2 lists the software stack; all of it is free for academic use.

Table 3.2: Software requirements

Software	Version / detail	Purpose
Unity Editor	6000.3.7f1 (Unity 6), Built-in Render Pipeline	Game engine, scene authoring, builds [15]
C# / .NET (Unity scripting)	Unity-bundled	All run-time logic (110 scripts)

Software	Version / detail	Purpose
OpenXR plug-in, Oculus XR	Unity packages	Headset runtime abstraction [16]
XR Interaction Toolkit	Unity package	Grabbing, sockets, locomotion, world-space UI [17]
Unity Input System	Unity package	Unified keyboard, mouse and controller input
TextMeshPro	Unity package	World-space and HUD text
Unity VideoPlayer	Built in	Recorded molecular videos (MP4)
Convai REST API	character/getResponse endpoint	Cloud conversational AI and voice [18]
Windows speech synthesis	Windows 10/11	Offline spoken replies
Git and GitHub	Any recent	Version control and team collaboration

3.4 Feasibility Study

Technical: Unity 6, OpenXR, the XR Interaction Toolkit and the Convai REST endpoint are mature and documented; pouring is computed from tilt and time instead of a fluid simulation, which is light enough for a standalone headset, and the final code compiles against Unity 6000.3.7f1 with zero errors. **Economic:** all software is free and assets were free; the only significant cost is an optional headset, since the desktop mode needs none, and a reusable virtual lab avoids recurring reagent and disposal costs. **Operational:** game-like controls, an onboarding card and a live objective line make the system easy to learn; teachers can use it for pre-lab preparation and quick tests with exported reports; dangerous reactions can be practised safely; and comfort settings (motion-safe camera defaults, typed questions) widen accessibility.

3.5 Cost Estimation

Table 3.3 gives an indicative cost estimate; prices are approximate and marked [ASSUMED].

Table 3.3: Cost estimation

Item	Quantity	Approx. cost (INR)	Remarks
Development laptops / PCs	3	0	Students' existing machines
Meta Quest 3S headset (VR testing)	1	30,000 [ASSUMED]	Optional for deployment; desktop mode needs none

Item	Quantity	Approx. cost (INR)	Remarks
Unity 6 Personal	3 seats	0	Free licence under revenue threshold
Visual Studio Community / VS Code	3	0	Free
Convai API (free tier)	1 account	0	Offline fallback when quota expires
3D models, textures, sounds	-	0	Free assets; audio cues synthesized in code
GitHub repository	1	0	Free
Internet and miscellaneous	-	2,000 [ASSUMED]	Package downloads, cloud AI use
Total (indicative)		32,000 [ASSUMED]	Dominated by the optional headset

3.6 Functional and Non-Functional Requirements

Table 3.4 lists the functional requirements. The key non-functional requirements are: interactive frame rates on a standalone headset; core features available without internet, microphone or AI credentials; comfort defaults (walking camera motion off, shake scalable to zero, room tone off); maintainability through data-driven recipes and one shared engine; portability through run-time desktop injection; and bounded, robust persistence (300-attempt JSON history, report-folder fallbacks).

Table 3.4: Functional requirements

ID	Requirement
FR1	The system shall present a 3D chemistry laboratory with grabbable glassware, reagents and a Bunsen burner.
FR2	The user shall select one of eight reactions from a world-space Reaction Book or with keys 1-8.
FR3	The system shall offer LEARN (recorded video or live 3D molecular view) and PERFORM for each reaction.
FR4	The system shall measure poured volumes, solid additions and heating time live and display them.
FR5	The system shall judge each experiment for wrong order, overdose, underdose and procedure errors against a recipe and tolerance.
FR6	The system shall display a chemistry-specific explanation of every failure and allow a bench reset (F5).
FR7	The AI assistant shall accept voice (push-to-talk) and typed questions and reply in voice and text.
FR8	The AI assistant shall receive the live experiment context and shall answer offline when the cloud is unavailable.
FR9	The system shall record every attempt (quantities, steps, outcome, AI Q&A) and persist up to 300 attempts.

ID	Requirement
FR10	The system shall score each successful attempt (accuracy %, 1-3 stars) and offer Guided, Standard and Expert levels.
FR11	The system shall show energy-profile, enthalpy and entropy graphs and let the user ask the AI to explain them.
FR12	The testing scene shall generate randomized practical and theory tasks with a timer and a chosen task count.
FR13	The testing scene shall maintain a persistent coin wallet with hints, extra time and skip purchases and ranks.
FR14	The system shall export a test report in HTML and CSV.
FR15	The application shall run in VR and, without a headset, with keyboard and mouse.

CHAPTER 4: PROJECT ANALYSIS & DESIGN

4.1 Introduction

This chapter models Atomix with UML use-case, sequence, state and class diagrams and with data-flow diagrams at levels 0, 1 and 2. All diagrams were drawn from the implemented code.

4.2 UML Diagrams

Use-case diagram. In Figure 4.1 the Student uses all eleven use cases, from selecting a reaction to downloading a test report. The Convai cloud (or its offline knowledge-base substitute) answers assistant questions, and the Instructor consumes exported reports.

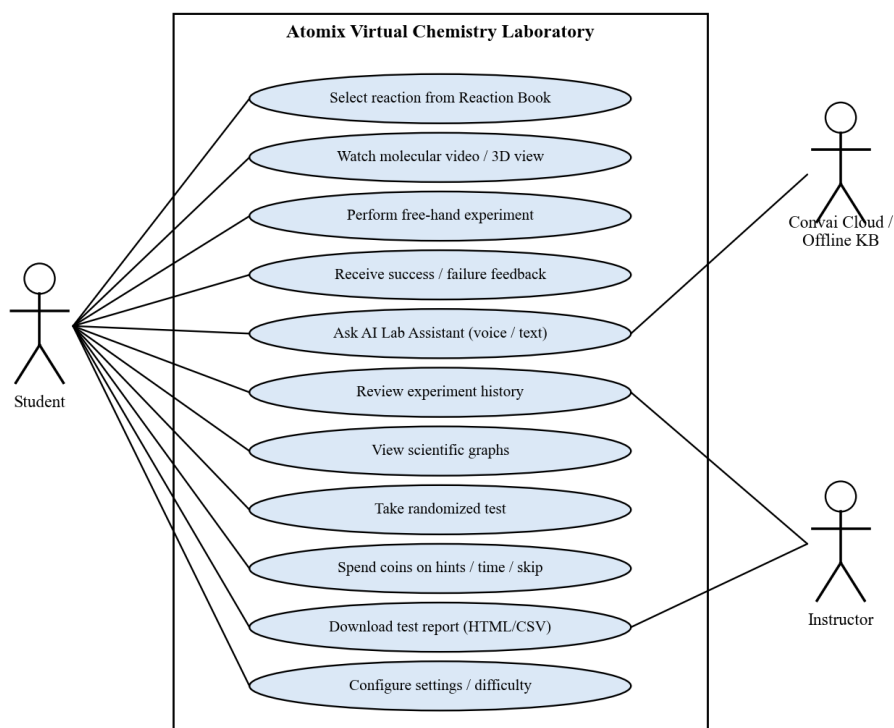


Figure 4.1: Use-case diagram of Atomix

Sequence diagram. Figure 4.2 shows an experiment: the Reaction Book calls `ControlReactions.SelectReaction(n)`, which activates reaction `n` and configures the shared `FreeHandReactionEngine` from its `ReactionDefinition`. Pours and additions are reported with `AddQuantity`; `Evaluate` runs every frame until a verdict, after which

effects, sound and the equation or failure reason are shown and the attempt is saved to the history manager.

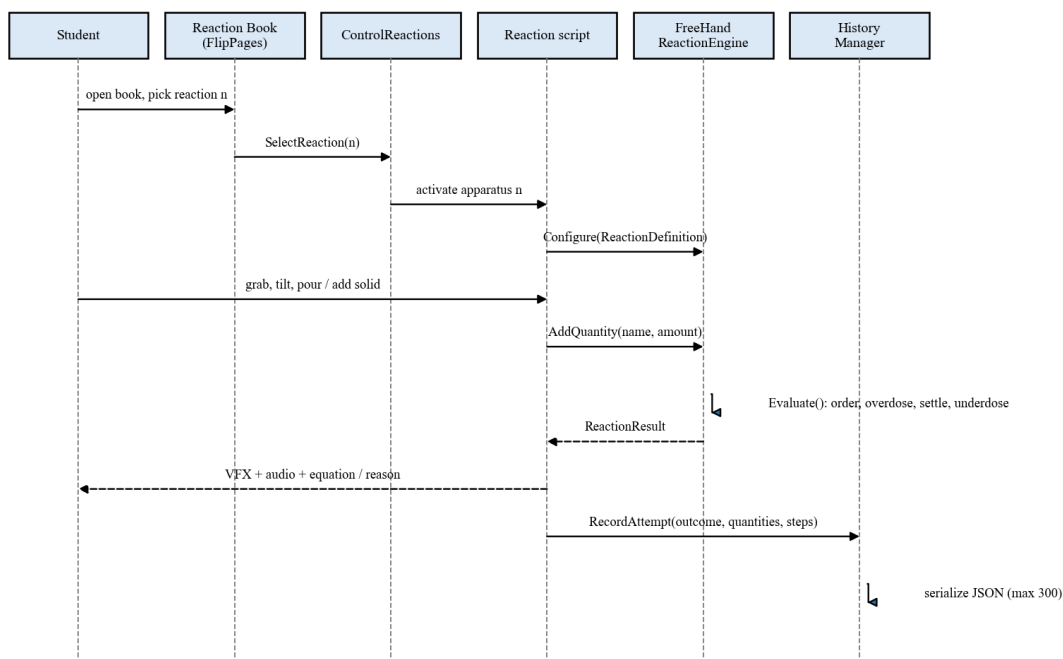


Figure 4.2: Sequence diagram - performing a free-hand experiment

State diagram. Figure 4.3 shows the engine's states. Overdose is checked every frame, so it fails immediately even mid-pour; once all reagents are in and pouring stops, a 1.5 s settle timer runs before underdose is judged, so a learner who pauses is never failed unfairly. Wrong order and heating-procedure errors are separate failures, and F5 resets every terminal state.

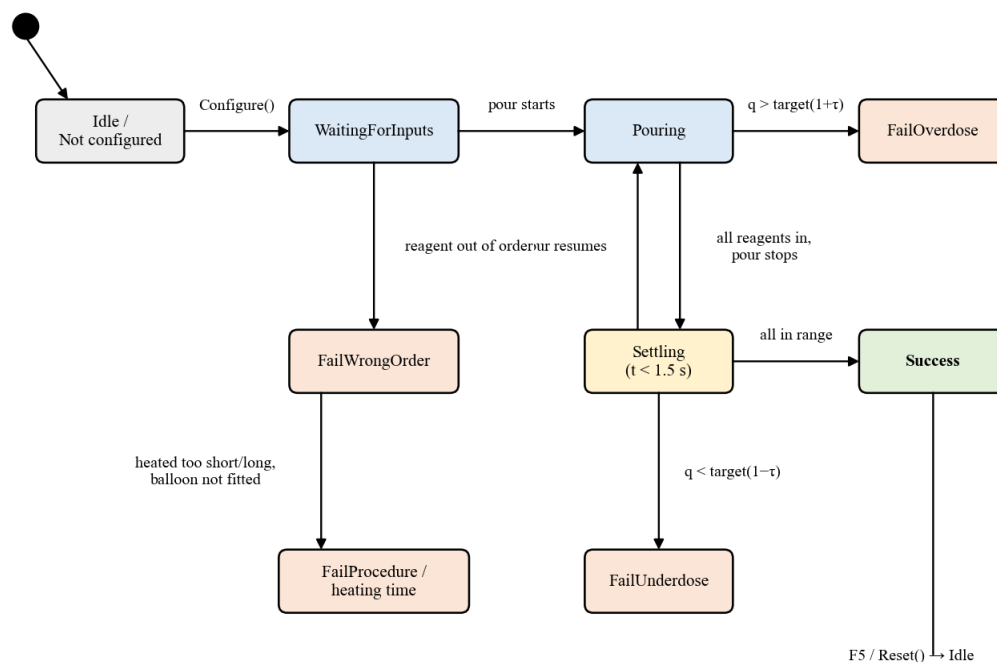


Figure 4.3: State diagram of the free-hand reaction engine

Data flow diagrams. At level 0 (Figure 4.4) Atomix exchanges input, feedback and speech with the student, questions and replies with the Convai cloud, and history and reports with the local file system. Level 1 (Figure 4.5) decomposes the system into input handling, reaction selection and learning, reaction evaluation, AI assistance, assessment and history, with recipe, PlayerPrefs and history stores. Level 2 (Figure 4.6) refines evaluation into reagent registration, quantity accumulation, order and overdose checks, settling, underdose and procedure checks, scoring and feedback.

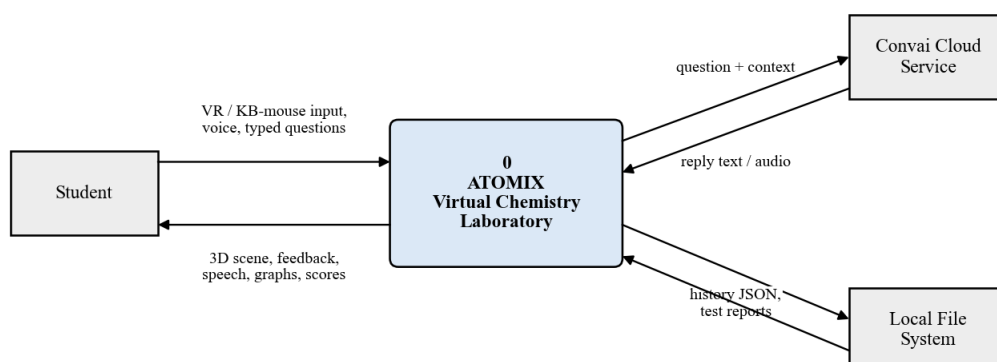


Figure 4.4: Level 0 DFD (context diagram)

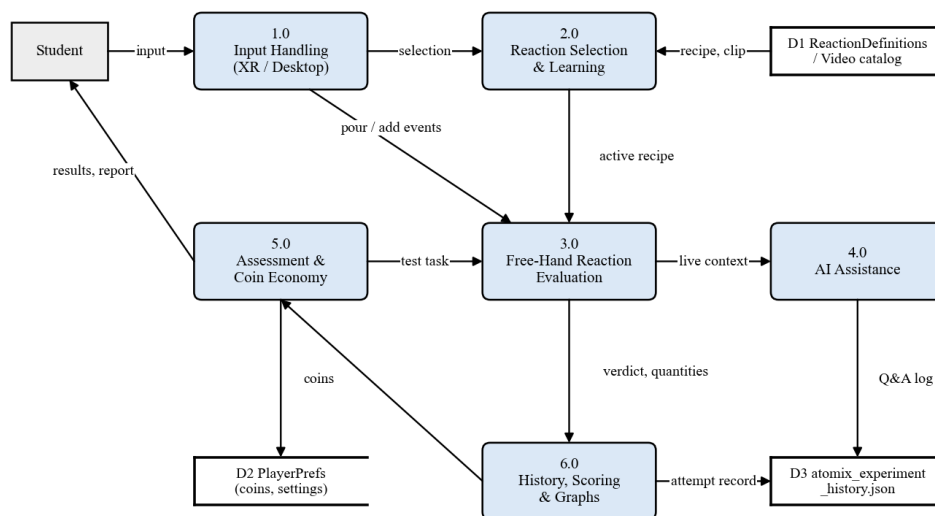


Figure 4.5: Level 1 DFD

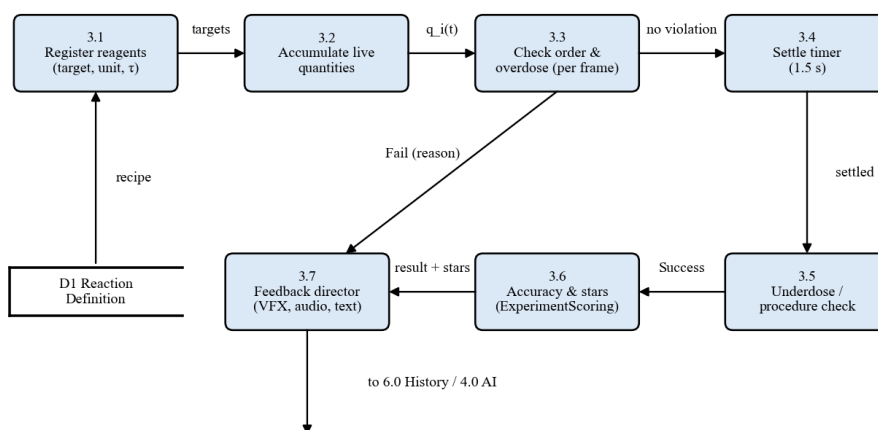


Figure 4.6: Level 2 DFD of the reaction-evaluation process

Class diagram. Figure 4.7 shows the core classes: ControlReactions selects among eight reaction scripts; a ReactionDefinition configures the engine; ExperimentScoring rates attempts; ExperimentHistoryManager persists them; the InLabAssistantController uses the Convai backend with the ChemistryKnowledgeBase as fallback; and AtomixCoinBank holds the economy.

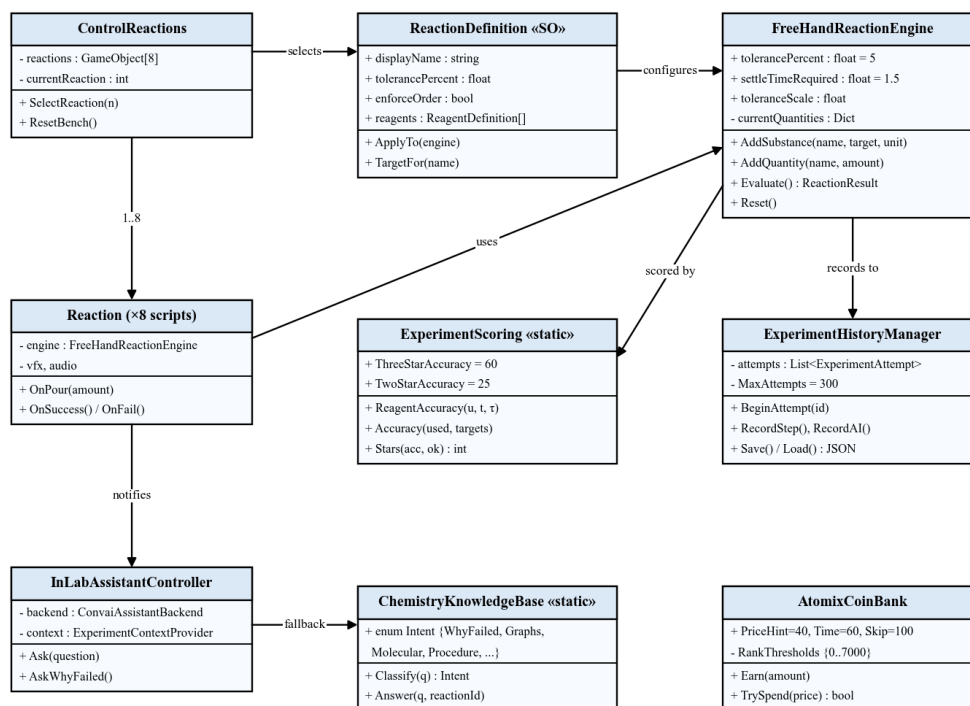


Figure 4.7: Class diagram of the core subsystems

4.3 Scene Architecture

Atomix uses four scenes (Table 4.1) so that each loads only the systems it needs.

Table 4.1: Scene architecture

Scene	Responsibility	Key systems
MainMenuScene	Entry point; mode selection; settings	MainMenu, world-space canvas, async loader
LabScene	Practice laboratory	Reaction Book, ControlReactions, eight reactions, FreeHandReactionEngine, history, graphs, in-lab AI, periodic table
TestingPhaseLab	Assessment	Randomize, TheoreticalTasksManager, CountdownTimer, ExamSession, AtomixCoinBank, ExamReportExporter
LabAssistantScene	Dedicated AI tutor with avatar	LabAssistantCharacter, InLabAssistantController, avatar animator

CHAPTER 5: METHODOLOGY

5.1 Introduction

Atomix was developed incrementally: a first prototype was extended through ten feature-and-fix iterations, each compiled against the real Unity assemblies and checked by automated tests before merging.

5.2 System Block Diagram

Figure 5.1 shows five layers. The **input layer** uses the XR Interaction Toolkit on a headset; otherwise DesktopBootstrap injects a first-person controller, crosshair and ray grabbing at run time, and LabTextInput captures typed questions while gating the keyboard. The **presentation layer** holds the Reaction Book, learning panel and HUD. The **simulation core** holds ControlReactions, the eight reaction scripts, the engine and the feedback director. The **intelligence layer** holds the assistant, scoring and graph catalog, and the **persistence layer** holds history, test session, coin bank and report exporter.

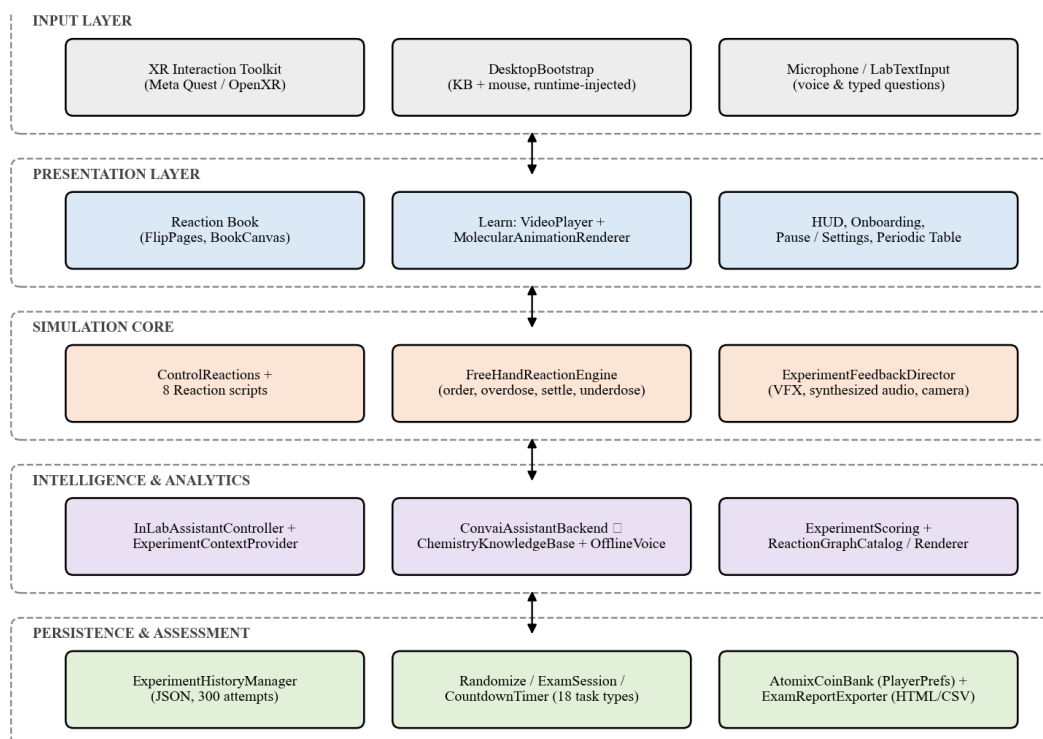


Figure 5.1: System block diagram of Atomix

5.3 Reaction Recipes and Evaluation Algorithm

Each experiment is a ReactionDefinition asset (Table 5.1). For reagent i with target T_i , tolerance τ_i and difficulty scale s (1.5 Guided, 1.0 Standard, 0.6 Expert), the accepted range is given by Equation (5.1).

Table 5.1: Reaction recipes used by the free-hand engine

#	Reaction	Reagents and targets (in order)	Tolerance τ
1	$2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$	Water 50 ml; Sodium 5 g	$\pm 5 \%$
2	$\text{H}_2\text{SO}_4 + \text{CuO} \rightarrow \text{CuSO}_4 + \text{H}_2\text{O}$	H_2SO_4 20 ml; CuO 8 g	$\pm 8 \%$
3	$\text{HCl} + \text{NaHCO}_3 \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	HCl 15 ml; NaHCO_3 12 g	$\pm 10 \%$
4	$2\text{K} + 2\text{H}_2\text{O} \rightarrow 2\text{KOH} + \text{H}_2$	Water 50 ml; Potassium 3 g	$\pm 5 \%$
5	$2\text{Al} + 3\text{I}_2 \rightarrow 2\text{AlI}_3$	Aluminium 5 g; Iodine 15 g; Water drops 2 ml	$\pm 10 \%$
6	$\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$	Water 30 ml; CaO 15 g	$\pm 8 \%$
7	$\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ (heat)	Heating 10 s (balloon fitted first)	$\pm 15 \%$
8	$2\text{FeSO}_4 \rightarrow \text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{SO}_3$ (heat)	Heating 10 s	$\pm 15 \%$

$$T_i(1 - s \cdot \tau_i/100) \leq q_i \leq T_i(1 + s \cdot \tau_i/100) \quad (5.1)$$

Liquid quantity is integrated from a tilt-dependent pour rate, solids increase in discrete additions and heating accumulates time over the flame. Every frame the engine:

1. Returns **FailWrongOrder** if a reagent is added before an earlier one.
2. Returns **FailOverdose** immediately if any q_i exceeds its upper bound.
3. Returns InProgress while pouring or while any reagent is missing, resetting the settle timer.
4. Otherwise accumulates the settle timer t_s (Equation 5.2) and waits until $t_s \geq 1.5$ s.
5. Returns **FailUnderdose** if any q_i is below range, or a procedure failure for wrong heating or (for CaCO_3) heating before fitting the balloon.
6. Otherwise returns **Success**, scores the attempt and records it.

$$t_s \leftarrow (t_s + \Delta t) \cdot [A(t) \wedge \neg P(t)] \quad (5.2)$$

where $A(t)$ is true when all reagents are present and $P(t)$ while any pour is active. Table 5.2 lists the verdicts and the feedback given.

Table 5.2: Engine verdicts and feedback

Verdict	Trigger	Feedback to the learner
Success	All quantities within range after settling; procedure correct	Reaction effect, sound, balanced equation, accuracy and stars, graphs unlocked
FailOverdose	Any $q_i > q_i^{\max}$ (checked every frame)	Reagent-specific chemistry reason, e.g. excess reagent pushes mixture away from the balanced equation
FailUnderdose	Any $q_i < q_i^{\min}$ after 1.5 s settling	Which reagent was short and its chemical consequence
FailWrongOrder	Reagent added before an earlier one	Why the order of addition matters for this reaction
Procedure failure	Heating too short/long; balloon not fitted (CaCO_3)	e.g. CO_2 escaped into the room instead of being collected

5.4 Accuracy and Star Model

Each reagent's accuracy falls linearly from 1 on target to 0 at the edge of its Standard range, and the attempt scores its worst reagent (Equations 5.3 and 5.4). Three stars need $A \geq 60$, two need $A \geq 25$, and any success earns at least one. The minimum prevents a careful measurement from hiding a careless one, and using the Standard range on every level keeps stars comparable. Figure 5.2 plots the model for H_2SO_4 in reaction 2 (accepted 18.4-21.6 ml).

$$a_i = \text{clamp}(1 - |q_i - T_i| / (T_i \cdot \tau_i / 100), 0, 1) \quad (5.3)$$

$$A = 100 \cdot \min_i a_i \quad (5.4)$$

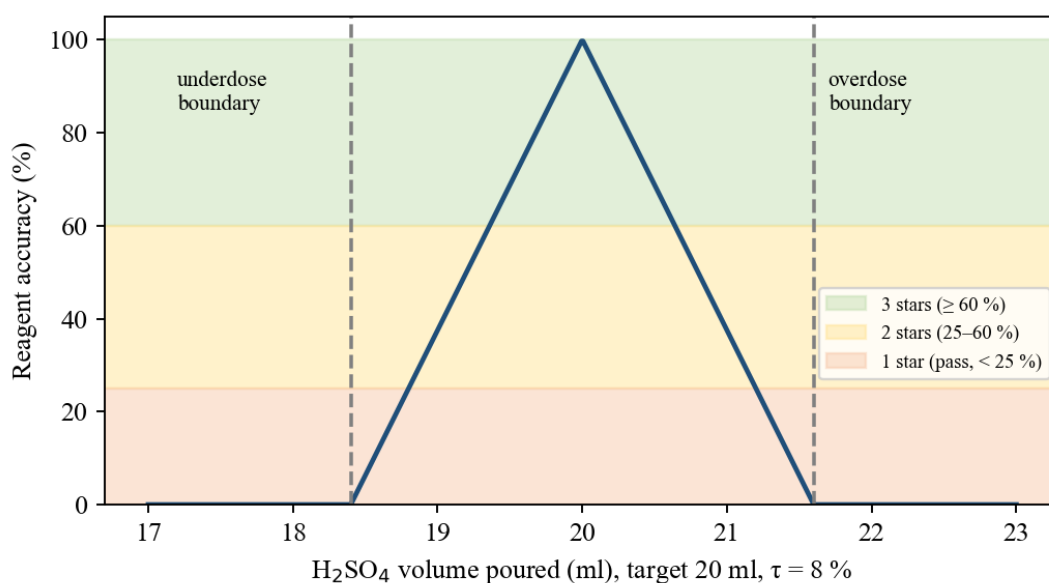


Figure 5.2: Reagent accuracy function and star bands for H_2SO_4 in reaction 2

5.5 Context-Aware AI with Offline Fallback

Questions arrive by push-to-talk (V), typing (Enter), a "why did it fail" key (Y) or Ask-AI buttons on graphs and animations. Each is sent with the reaction, quantities, targets, verdict and attempt counts (Figure 5.3). With credentials and network, Convai's character endpoint returns text and speech; otherwise the ChemistryKnowledgeBase classifies the question by keyword scores $S_k(q)$ (Equation 5.5) into intents such as WhyFailed, Quantities, Procedure, Molecular, Graphs, Calculation, Safety and Progress, and answers from game data - e.g., "you poured 27.4 ml where the equation needs 20.0, a 37 % overdose" - using Windows speech.

$$k^* = \operatorname{argmax}_k S_k(q); \text{ Overview if all } S_k(q) = 0 \quad (5.5)$$

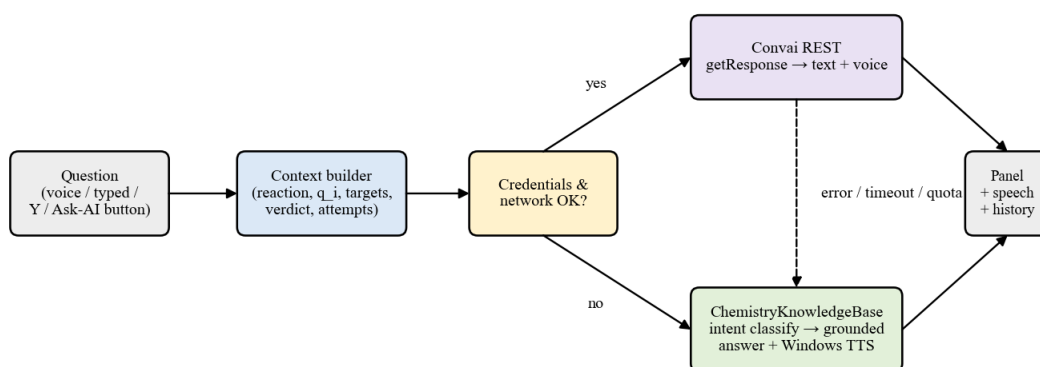


Figure 5.3: Dual-path routing of AI questions

5.6 Molecular, Thermodynamic and Assessment Models

The live 3D view is authored in MolecularSceneCatalog: a thinning red bond is breaking, a thickening green bond is forming and a yellow dot is an electron transferred, shown only for the redox reactions. The graph catalog stores ΔH , E_a and ΔS per reaction and derives Gibbs energy, crossover temperature and reverse barrier (Equations 5.6-5.8; ΔH , ΔG , E_a in kJ/mol, ΔS in J/mol·K).

$$\Delta G(T) = \Delta H - T \cdot \Delta S / 1000 \quad (5.6)$$

$$T^* = 1000 \cdot \Delta H / \Delta S \quad (5.7)$$

$$E_{a,rev} = E_a - \Delta H \quad (5.8)$$

The test scene draws 3-18 tasks from 8 practical and 10 theory items; practical tasks use the same engine with hidden targets. Coins persist across sessions; the balance is B

$= \max(0, L - S)$ for lifetime earned L and spent S , and $\text{rank} = \max\{r : L \geq \theta_r\}$ with $\theta = \{0, 500, 1500, 3500, 7000\}$. Table 5.3 lists the parameters.

Table 5.3: Coin-economy parameters

Parameter	Value
Earning per task	5-100 coins, scaled by time left
Bonuses	First-ever clear +25; 3-streak +30; 5-streak +60; perfect unaided run +50
Purchases	Hint 40 (F2); +30 seconds 60 (F3); skip task 100 (F4)
Ranks	Apprentice (0), Lab Technician (500), Chemist (1500), Senior Chemist (3500), Lab Master (7000)
Hint content	Procedure only, never the quantity - the amount is what the test measures

CHAPTER 6: IMPLEMENTATION DETAILS

6.1 Introduction

Atomix is implemented in C# on Unity 6000.3.7f1 (Built-in Render Pipeline): 110 scripts totalling 39,601 lines, four scenes, eight recipe assets and seven molecular MP4 clips. Placeholders marked [INSERT SCREENSHOT] must be replaced with captures from the final build.

6.2 System Implementation

Laboratory. The lab (Figure 6.1) is built from free 3D models of benches, a sink, blackboard, periodic-table poster and glassware, with ambient occlusion and a box-projected reflection probe. The capture shows the FeSO_4 experiment set up while the assistant explains how to light the burner.



Figure 6.1: The Atomix virtual laboratory with the FeSO_4 experiment set up

Reaction Book and learning. The world-space book (Figure 6.2) lists reactions by category with LEARN / PERFORM and a difficulty selector; keys 1-8 select directly. LEARN plays a video through ReactionLearningVideoCatalog or, via 3D VIEW, the live animation rendered by MolecularAnimationRenderer (Figure 6.3).

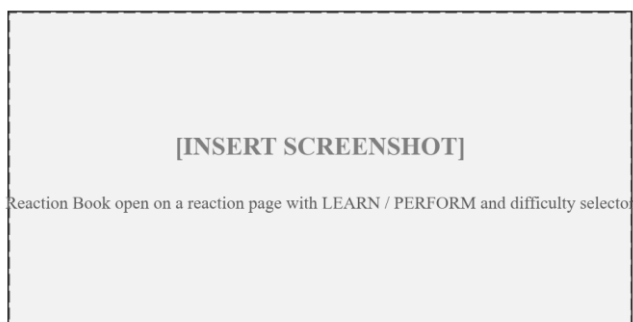


Figure 6.2: Reaction Book with LEARN / PERFORM and difficulty selection

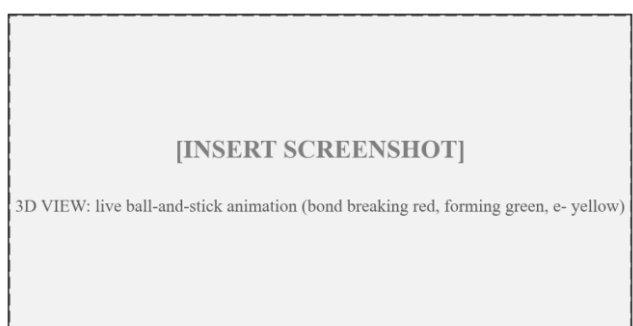


Figure 6.3: Live 3D molecular animation

Experimentation, history and graphs. PERFORM enables only that reaction's apparatus. Tilting a held vessel past a threshold emits a pour stream and raises the receiver's quantity; a tooltip and tracker show live amounts. Verdicts trigger synthesized success or gentle failure tones, camera shake and a result or failure card; F5 resets the bench. Tab opens the history (filters, expandable steps, used versus target, AI log) and F the energy-profile, enthalpy and entropy graphs with "Ask AI to Explain".

AI assistant. The panel sits at the right edge (Figure 6.4). LabTextInput takes over the keyboard while typing and is refused mid-pour; ConvaiAssistantBackend posts text and context with UnityWebRequest; OfflineVoice speaks offline answers.



Figure 6.4: In-lab AI assistant answering a typed question

Testing scene. ExamSetupUI asks for the task count; Randomize and TheoreticalTasksManager present practical and multiple-choice tasks; ExamCoinHud shows the wallet and the F2/F3/F4 shop; ExamSession records every task; and the report card exports HTML and CSV reports (Figure 6.5).



Figure 6.5: Testing scene report card and exported HTML report

6.3 Desktop and VR Modes

DesktopBootstrap uses RuntimeInitializeOnLoadMethod to add, when no XR device is active, a FirstPersonController (W/A/S/D, mouse look, sprint, fly), a crosshair, ray-based grabbing, a LabBoundary clamp and a DesktopObjectSettler that lowers floating glassware onto the bench. Held objects are rotated with Q/E, tilted with Z/X and rolled with C. In VR, an XR Origin with grab interactables, ray-interactable canvases, locomotion and Oculus hand models is used; because amounts derive from tilt and time, the same reaction scripts serve both modes. Table 6.1 lists the key scripts.

Table 6.1: Key scripts and responsibilities

Script	Responsibility
ControlReactions.cs	Reaction selection and laboratory set-up
FreeHandReactionEngine.cs (1,264 lines)	Shared quantity, order, overdose, underdose and settle logic
ReactionDefinition.cs	Data-driven recipe asset; configures the engine
ExperimentScoring.cs	Accuracy percentage, stars and difficulty scales
ExperimentHistoryManager.cs / ExperimentHistoryUI.cs	Attempt persistence and browsing
InLabAssistantController.cs / ExperimentContextProvider.cs	In-lab AI and context building
ConvaiAssistantBackend.cs	REST calls to the Convai character endpoint
ChemistryKnowledgeBase.cs / OfflineVoice.cs	Offline intent-based answers and Windows speech
MolecularSceneCatalog.cs / MolecularAnimationRenderer.cs	Code-authored molecular stages and live rendering

Script	Responsibility
ReactionGraphCatalog.cs / ReactionGraphRenderer.cs / ReactionGraphUI.cs	Thermodynamic data and graphs
Randomize.cs / ExamSession.cs / CountdownTimer.cs	Randomized timed assessment
AtomixCoinBank.cs / ExamCoinHud.cs / ExamReportExporter.cs	Wallet, shop, HTML/CSV report
DesktopBootstrap.cs / FirstPersonController.cs / LabTextInput.cs	Desktop mode and typed-input gate

CHAPTER 7: RESULT ANALYSIS

7.1 Introduction

Atomix is a functional prototype verified by compilation and by automated checks that run the shipped logic. No controlled learning study, usability survey or on-device frame-rate measurement has yet been carried out, so none is reported; placeholders mark where measured data should be added.

7.2 Performance Measures

Engine-independent logic was extracted verbatim and executed with a time shim. Table 7.1 and Figure 7.1 show that every suite passed in full; the outcome matrix drove each test-scene experiment through correct amounts, overdose, underdose, reversed order and a mid-procedure pause, and confirmed that targets are never revealed in the test.

Table 7.1: Build and verification summary

Verification item	Result
Compilation against Unity 6000.3.7f1 assemblies	0 errors; 68 pre-existing warnings (CS0649/CS0414 in original scripts); 0 new warnings
Reaction-engine outcome matrix (8 test-scene experiments)	111 / 111 checks passed
Regression suite after molecular animation and offline AI	167 / 167 checks passed
Regression suite after coin economy and exam report	226 / 226 checks passed
Synthesised-audio suite	22 / 22 checks passed
Full Play-mode regression on headset and desktop	[INSERT: results of final Play-mode pass]

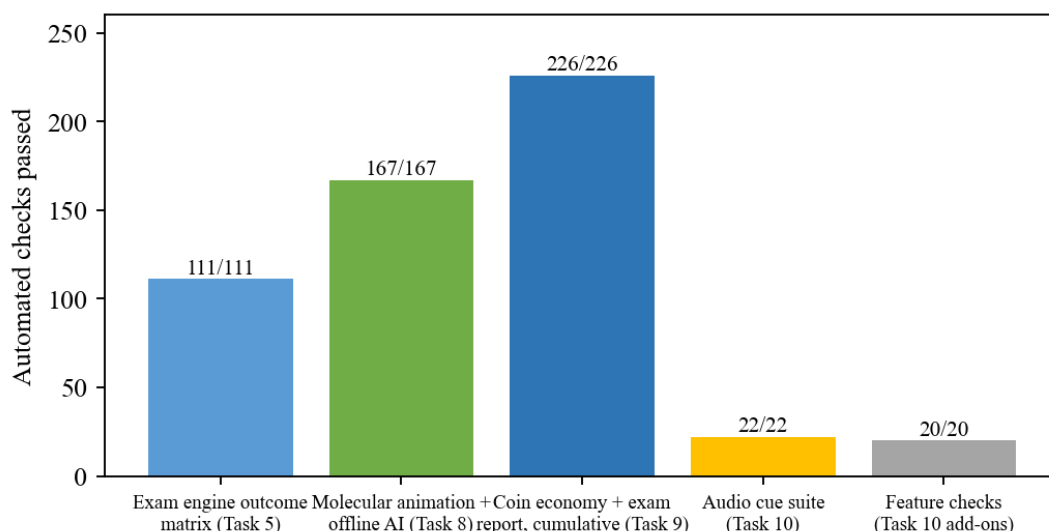


Figure 7.1: Automated verification checks passed per suite

Table 7.2 lists the thermodynamic data used by the graphs with ΔG at 298.15 K from Equation (5.6), and Figure 7.2 plots ΔH and E_a . Only the two thermal decompositions have positive ΔG at room temperature, which is why they are the experiments needing the burner (crossover about 1,110 K and 901 K). Reaction 3 is endothermic yet spontaneous through its entropy gain, and the alkali-metal reactions combine tiny barriers with large exothermicity, matching their violence in water. Other content metrics: 18 task types, 8/8 live animations, 13 achievements and a 118-element periodic table. [INSERT: frame rate on Quest and desktop; Convai versus offline response latency; SUS and pre/post-test scores.]

Table 7.2: Thermodynamic data per reaction and derived Gibbs energy

#	Type	ΔH (kJ/mol)	E_a (kJ/mol)	ΔS (J/mol·K)	ΔG_{298} (kJ/mol)	Spontaneous at 298 K
1	Exothermic	-368.6	10	-15.5	-364.0	Yes
2	Exothermic	-63.7	48	-43.0	-50.9	Yes
3	Endothermic	+31.4	45	+241.0	-40.5	Yes (entropy-driven)
4	Exothermic	-393.2	6	+44.7	-406.5	Yes
5	Exothermic	-627.6	85	-86.9	-601.7	Yes
6	Exothermic	-65.2	46	-26.3	-57.4	Yes
7	Endothermic	+178.3	185	+160.7	+130.4	No (needs heating)
8	Endothermic	+340.1	365	+377.4	+227.6	No (needs heating)

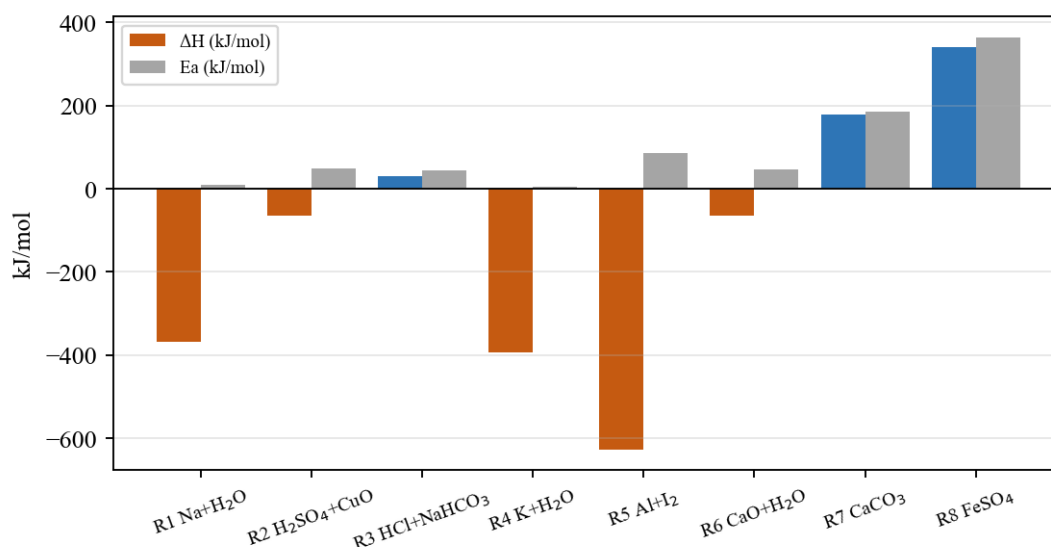


Figure 7.2: Enthalpy change and activation energy of the eight reactions

Table 7.3 traces the functional requirements of Table 3.4 to their implementation and evidence.

Table 7.3: Requirement traceability

ID	Implemented by	Status / evidence
FR1	LabScene, 3D models, grab interactables	Implemented; Figure 6.1
FR2	FlipPages, BookCanvasManager, keys 1-8	Implemented
FR3	ReactionLearningController, MolecularAnimationRenderer	Implemented; 8/8 animations, 167/167 suite
FR4, FR5	FreeHandReactionEngine, ReactionDefinition	Implemented; 111/111 outcome matrix
FR6	ExperimentFeedbackDirector, LabRetryController (F5)	Implemented
FR7, FR8	InLabAssistantController, ConvaiAssistantBackend, ChemistryKnowledgeBase	Implemented; offline path verified in 167/167 suite
FR9, FR10	ExperimentHistoryManager, ExperimentScoring	Implemented
FR11	ReactionGraphCatalog / Renderer / UI	Implemented
FR12- FR14	Randomize, ExamSession, AtomixCoinBank, ExamReportExporter	Implemented; 226/226 suite
FR15	XR rig + DesktopBootstrap	Implemented; headset Play-mode pass [INSERT]

7.3 Screenshots

The main screens are shown in Figures 6.1 to 6.5.

7.4 Comparative Analysis

Table 7.4 compares Atomix functionally with representative systems from Chapter 2 (their numerical results used different populations and are not directly comparable). Atomix is the only reviewed system combining quantity-aware judging, molecular views, a live-context AI that works offline, attempt history, thermodynamics and assessment across VR and desktop; its main weakness is the absence, so far, of a controlled evaluation like [6]. Known limitations include online-only cloud AI, Windows-only offline speech, per-machine coins and a committed API key that must be rotated.

Table 7.4: Comparison with existing systems

Feature	Virtual labs [1],[12]	VR systems [4],[13]	Molecular VR [5]	AI chatbots [7],[14]	Atomix
Platform	Desktop / web	VR	Projection CAVE	Web / messaging	Meta Quest + Windows desktop
Experiment judging	Step sequence	Scripted	None	None	Order, overdose, underdose, heating, with tolerance
Difficulty levels	Rare	No	No	No	Guided / Standard / Expert
Molecular view	Static images	Partial	Yes	No	Video + live animation for 8/8 reactions
AI help	No	No	No	Yes, generic	Context-aware, voice + text, offline fallback
Progress record	Scores	No	No	Chat log	300-attempt history with steps, accuracy, AI log
Thermodynamics	No	No	No	No	ΔH , E_a , ΔS , ΔG graphs with AI explanation
Assessment	Quizzes	No	No	No	18 task types, coins, HTML/CSV report

CHAPTER 8: CONCLUSION & FUTURE SCOPE

8.1 Conclusion

Atomix unifies quantity-aware free-hand experimentation, molecular visualization, a context-aware AI assistant that works offline, attempt history, thermodynamic graphs and randomized assessment in one Unity 6 application for VR and desktop. Its data-driven engine judges order, overdose, underdose and procedure against per-reaction tolerances scaled by difficulty, and its minimum-reagent model rates attempts fairly. The 110-script, 39,601-line implementation compiles without errors and passed all automated checks (111/111, 167/167, 226/226, 22/22), and every functional requirement was implemented. These results show technical and functional viability; the educational effect remains to be measured.

8.2 Future Scope

- A controlled pre-test/post-test study with usability questionnaires.
- More experiments (titration, organic, electrochemistry) as new recipe assets.
- A teacher dashboard with accounts, synced history and leaderboards.
- An on-device language model grounded on the knowledge base; offline speech beyond Windows.
- Multiplayer benches, mixed-reality passthrough, WebGL/tablet builds and Hindi/Marathi localisation.

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